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Techno-economic feasibility analysis of a hybrid off grid AC-DC microgrid to support a health clinic

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ABSTRACT

Techno-economic analysis of off-grid hybrid AC-DC microgrids (HMGs) in desert areas has primarily focused on meeting the load demands in residential, household, domestic, and agricultural applications, while the healthcare sector has been comparatively less considered. These analyses contribute to the development of effective financial techniques. This paper presents a techno-economic design for an HMG that combines AC and DC elements such as diesel generators (DG), photovoltaic (PV), wind turbines (WT), batteries (BAT), and power converters (Conv) to satisfy the power demand of a rural health clinic with a daily load of 110 kWh, situated in the desert area of Nubian in Aswan, Egypt. The optimization of HMG is performed through HOMER Pro based on hourly wind speed, solar irradiance, and clinic load data to evaluate the cost of electricity, carbon emissions, loss of power supply probability, and renewable fraction. The results are compared across four different HMG combinations: PV/WT/BAT/Conv, DG/PV/BAT/Conv, DG/PV/WT/BAT/Conv, and WT/DG. The simulation outcomes indicate that the system incorporating the WT/DG/PV/BAT/Conv provides the most efficient techno-economic solution for meeting the clinic's power demand. The optimal configuration includes 4 kW of DG, 10 kW of WT, 9 kW of PV, 36 batteries, and 16 kW of power converters. This system achieves the lowest 51.76 k\$ of net present cost and 0.107 \$/kWh of cost of electricity, 3051 kg/year of CO₂ emissions, and a significant renewable contribution of 90.7 %. Furthermore, the sensitivity assessment verifies that the system costs are greatly affected by factors including solar radiation, wind speed, discount rate, and diesel cost.

1. Introduction

1.1. Background

The government of Egypt set a goal under the integrated and sustainable energy strategy (ISES-2035) policy to improve renewable power generation to 42 % of its total electrical power production. A further objective of the plan is to cut energy use by 18 % using energy efficiency and demand management techniques [1]. The strategic geographical position of Egypt and the government's effort to diversify electrical power generation toward renewable energy sources (RES) create a favorable environment for sustainable energy development. The country benefits from substantial solar potential with an annual sunlight range of

2900 to 3200 hours (hrs). Additionally, its desert areas receive monthly an average of 7 kW hours per square meter per day (kWh/m²/day) of global horizontal irradiance (GHI), which makes them suitable for the production of photovoltaic (PV) systems. Furthermore, some areas have 10 meters per second (m/s) average wind speeds, which are perfect for power generation from wind turbines (WT) [2]. However, the unpredictable and intermittent nature of weather conditions limits the use of PV and WT systems [3]. Therefore, a battery storage system (BSS) combined with conventional and renewable technologies in the hybrid an off-grid AC-DC microgrid (HMG) has been determined as an effective way to improve energy security, resilience, and sustainability. This approach is particularly beneficial in distant, remote, or island

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locations where grid connections are expensive and complicated. In the HMG system, feasibility and optimal analysis of PV, WT, and BSS are very important. Significant unmet load demand and operational failures may result from under sizing. Whereas, oversizing can result in excessive energy production and high system costs [4]. Therefore, finding a techno-economic approach is important for achieving excellent dependability at the lowest possible cost and improving system availability and reliability. As a result, considerable research has been devoted to assessing the techno-economic assessment of the HMG based on feasibility and optimal size parameters. This has involved designing numerous microgrid (MG) schemes to supply power to a wide range of loads. In this techno-economic analysis, MATLAB and hybrid optimization of multiple energy resources (HOMER) tools are used to achieve the economic costs for cost of electricity (COE) and net present cost (NPC), improving the renewable fraction (RF) and loss of power supply probability (LPSP), and providing a reliable electrical supply. This study is motivated by the growing need for sustainable, affordable, and dependable energy solutions in remote and isolated areas where grid access is limited or unavailable. Integrating various RES, including PV, and WT with the diesel generator (DG), along with BSS provides a promising method to achieve energy independence and environmental sustainability. However, determining optimal sizing and operation of HMG remains complex due to the variability of RES and cost considerations. Therefore, this research focuses on developing and optimizing HMG that improves the usage of RES, reduces cost and emissions, and ensures a dependable power supply for real world applications. Specifically, it aims to meet the load demand in various sectors, such as providing power to health clinics in the Nubian (Aswan) region.

1.2. Literature review

Recent research has made a significant advancement in the design of techno-economic analysis techniques for off grid HMG in Egypt. In [5], an off grid HMG powered by WT and PV is proposed to supply electricity to digestion plants located at the Tiba site. This analysis report provides 0.1158 dollars per kilowatt hour (\$/kWh) of COE and 6894.7 thousand dollars (k\$) of NPC achieving a 100 percentage (%) of RF. Similarly, another off grid alternating current micro grid (ACMG) is designed to fulfill the electricity demand of an international school situated in Cairo [6]. The optimal system configuration derived from HOMER includes 180 kW (kW) WT, 50 kW capacities for electrolyzer, fuel cell (FC) and WT, 200 kW of PV, and a 686 kWh (kWh) battery. This system results in COE of 0.153 \$/kWh and NPC of 1775.3 k\$. In [7], an HMG consisting of PV, biomass, battery (BAT), and WT is developed to satisfy the Abu Minqar load demand by using HOMER Pro simulation tool. This HMG scheme provides NPC of 3483.51 k\$, COE of 0.118 \$/kWh, and carbon dioxide (CO₂) emissions of 28,516 tons per year (Ton/yr). The irrigation electrical power requirement in Sadat (Egypt) is satisfied by employing the HMG scheme consisting of PV, BAT, WT, and power converter (Conv) [8]. This work uses the HOMER tool to determine the NPC, RF, and COE. The outcomes show that the design system provides COE of 0.119 \$/kWh, NPC of 194.017 \$, and RF of 100 %. Similarly techno-economic analysis of numerous WT, DG, hybrid BSS, and PV schemes has been studied to determine the optimal NPC through HOMER tool and meet the power demand of Nubian (a small village in Aswan) [9]. According to the analysis, the most economical option includes PV panels (with a vertical axis PV tracking mechanism), WT, and lithium-ion batteries. This system attains a COE of 0.255 \$/kWh and an NPC of 853.63 k\$. It also produces a respectable 64,018.15 kilogram (kg) of greenhouse gas emissions annually. A PV, DG, WT, Conv, and BAT scheme was proposed to meet the power demand of desalination plants in Egypt [10], in which HOMER was employed to evaluate the techno-economic. The configuration of desalination facilities in Egypt generates an RF of 93.1 %, a COE of 0.107 \$/kWh, and an NPC of 502.66 k\$. In a similar study, the power demand of a large-scale reverse osmosis desalination plant (RODP) in Egypt was satisfied by using

the PV, DG, WT, BAT, and Conv HMG setup [11]. The techno-economic assessment results of this configuration show that the optimal configuration includes 25 of WTs, 451 kW of PV, 250 kW of DG, 352 BATs, and 358 kW of Conv. This arrangement obtains NPC of 3120 k\$ and a COE of 0.164 \$/kWh. Moreover, the load demand of RODP in Baltim was met by proposing the HMG configuration [12]. In this model, numerous HMG alternatives were developed and investigated based on the fuzzy multicriteria decision technique. Among ten viable options, the result indicates that the optimal system comprises of five 20-kW WT, a 100 kW of DG, 328-kW of PV array, 235-kW Con, and 112 batteries. This configuration demonstrates the most economic performance with the optimal NPC (1048.04 k\$) and COE (0.101\$/kWh) values, and 95.55 % of RF. To achieve reduced NPC values for El Kharga Oasis (desert location in Egypt), an improved arithmetic optimization technique is proposed in [13]. The study results indicate that the developed HMG system that incorporates DG, PV, WT, and batteries demonstrates a significant level of cost effectiveness. This system offers a competitive COE of 0.2309 \$/kWh. Similar location have been used in [14] to assess the feasibility analysis of HMG system. This optimal size and feasibility assessment verifies that the DG, WT, PV, BAT and Conv based HMG is the best economical option. This HMG setup provides NPC at 103.43 k\$ and a COE at 0.108 \$/kWh with a high RF of 90.1 %. In [15], the authors have been introduced a systematic approach for optimal sizing and conducting sensitivity analysis of an independent MG system to satisfy the electrical power and heat demands of an urban household in Safaga (Egypt). Based on their investigation, among several design options, the most effective techno-economic design is provided by a 17.2-kW of PV, 70 BSS, a single 10-kW of WT, a 5-kW of FC, and a 13.7-kW inverter. Additionally, the analysis reveals that the NPC is highly sensitive to load profile, whereas variations in solar and wind energy have minimal impact on future investment. This configuration results in NPC of 1.23 k\$, RF of 42.6 %, and COE of 0.150 \$/kWh. According to [16], an HMG scheme combining PV, DG, WT, BAT, and Conv is feasible and optimal for meeting the electrical power load requirement in a community of Marsa-Matruh. HOMER pro result identified that the 41.2 kW of PV, 15 kW DG, 6 units of 2.625 kW of WT, and 23.92 kWh of lead-acid batteries with 92 units is the optimal capacity of most cost effective HMG setup. This system provides 351.22 k\$ of NPC, 57 % of RF, and COE of 0.2262 \$/kWh. The generalized mathematical model is introduced in [2] to identify the optimal sizes of WT, PV, and BSS in remote area of Sinai (Egypt). The optimal system determined by HOMER has 0.118 \$/kWh of COE, while grey wolf optimization (GWO) achieved approximately COE at 17 %. Additionally, GWO provided better convergence and superior statistical performance compared to the particle swarm optimization (PSO) method. The optimization technique based on the COE and reliability of the energy supply model of PV, FC, and WT is proposed to satisfy the electricity demand in a remote area of Beni-Suef [17]. In this method, multiple configurations are discussed and compared in detail in terms of optimal size analysis. Simulation outcomes verified that the designed firefly algorithm (FA) provides the most economical optimal size and offers NPC of 2.69 k\$ and COE of 0.45 \$/kWh. In Qusier [18], the HMG combining PV, WT, DG, Con, and BAT is developed to meet the household load demand. The optimal configuration consists of two 10 kW of WTs, 15 kW diesel generator, 20 kW of PV, 40 kW power converter, and 80 batteries. This scheme results in an NPC of 169 K\$, RF of 82.4 %, COE of 0.140 \$/kWh.

Furthermore, a novel optimization method called the mayfly optimization algorithm (MOA) has been applied to develop a small standalone MG to fulfill the power requirements of a remote site in Al-Aqaba, which has a peak demand of 420 kW [19]. PV, FC, a hydrogen tank (HT), an electrolyzer, a Conv, and biomass are all part of the established MG system. The most effective approach for this system was found by MOA, resulting in an NPC of 6170.13 k\$ and a COE of 0.2106533 \$/kWh. Similarly, the most recent dynamic control Cuckoo search (DCCS) optimizer is designed for the remote area of Al-Kharjah to improve energy management and provide best optimal size [20]. This

HMG system contains PV, WT, DG, and BSS components. The simulation outcome verifies that DCCS delivers a promising solution with the minimum COE at 0.1990 \$/kWh and NPC of 5798.7 k\$. In [21], techno-economic evaluation of an HMG system is performed in East Owienat area of Egypt, which utilizes PV, WT, BSS, microturbines, and HT. The study investigates the effects of the thermal source on CO₂ emissions and NPC of the most suitable configuration. This system provides NPC of 4228.11 k\$. Similarly, PV panels, WT, DG, BSS, and HT based MG scheme is suggested for the case study of residential communities in Baris [22]. This scheme utilizes an enhanced optimization algorithm known as Quadratic interpolation-based artificial rabbits optimization (QIARO), resulting in an NPC of 1852.42 k\$. A techno-economic study of an off-grid PV and biogas (BG) setup, as detailed in [23], was conducted to supply electricity to a household in a rural community in Sohag Al Gadida. NPC, COE, and RF were evaluated in the analysis using HOMER software. The outcomes showed that the hybrid PV, BG is the most efficient, meeting local power requirements at the lowest possible COE (0.184 \$/kWh), NPC of 11.026 k\$, and high RF of 99.9 %. An optimal HMG with PV panels, WT, and BSS is the best option for powering a desalination unit on a farm in Noubarya, Egypt [24]. This desalination unit requires approximately 105.6 kWh/day, with a rated power of 6.6 kW. Using the genetic algorithm (GA) in iHOGA software, several possible configurations were examined. According to the analysis, the proposed setup has the lowest COE (0.15 \$/kWh) and NPC (137.01 k\$). In a case study conducted by the authors in [25], the optimal isolated MG design with PV, biomass, WT, and a hydrogen energy storage system (HESS) was developed for the Ataka area. PSO, GWO, and Harris Hawks Optimization (HBO) methods were employed to determine the optimal size. According to the findings, the optimum design with a minimum COE of 0.2750 \$/kWh and an NPC of 8055.05 k\$ was achieved using the HBO algorithm. In [26] authors determined the optimal sizing of HMG based on algorithm of Colony Optimization (ACO). To identify the best configuration, the results are compared with those obtained using GA, HOMER, and PSO. The Kharga, Qussair, and Saint Katherine distant Egyptian locations with different meteorological conditions are selected for the HMG system optimization. The findings show that the HMG system's combined optimal design more successfully satisfies load demands while abiding by all constraints. In an isolated region of Marsa-Matruh, the study evaluated the techno-economic arrangement

and viability of a hybrid backup system integrated into an MG with vehicle-to-grid [27], in which BSS, electric vehicle battery (EV), and the HESS are employed as backup power sources. Various operating setups of backup power sources were implemented and carefully examined, producing comprehensive techno-economic outcomes. The snake optimizer (SO) and reptile search algorithm (RSA) were surpassed by the AVOA algorithm. At about 13.06 k\$ of NPC for the day, the hybrid HESS, EV, and BSS backup system optimized by AVOA demonstrated remarkable cost-effectiveness. In [28], the HMG system in Farafra site is designed through the enhanced Archimedes optimization algorithm (IAOA). Three different configurations for the MG based on PV panels, WT, BSS, and DG are explored. Outcome results are compared with numerous well-known algorithms, including the original Archimedes Optimization Algorithm (AOA), the Artificial Electric Field Algorithm (AEFA), the Equilibrium Optimizer (EO), the GWO, and the HHO algorithm. The results demonstrate the higher effectiveness of the suggested algorithm over the other methods and confirm its efficiency in solving the design difficulties. PV panels, WT, DG, and BSS constitute the optimum HMG, which has an NPC of 187.18 k\$, COE of 0.213 \$/kWh, and a RF of 90 %. Effective operational planning is very important for enhancing the performance of HMG systems. To achieve this, an optimal sizing method for the HMG, which includes PV, biomass, WT, batteries, and Conv components, is proposed to improve the sustainability and cost efficiency in urban sites [29]. This system provides an NPC of 107 k\$, and COE of 0.118 \$/kWh. In [30], an innovative framework combining the Optimized Artificial Neural Networks (O-ANN) and the Derivative Log Sigmoid-Woodpecker Mating Algorithm (DLS-WMA) is developed to merge cogeneration power plants with PV systems. This framework proves effective in maintaining a stable power output despite the intermittent nature of RES, highlighting its practical applicability and potential for broad use in industrial applications. It enables up to 25 % higher power production from RES. The optimal power flow (OPF) scheme based on average load ability index [31], along with the optimal placement of voltage source converter based high voltage direct current system (VSC-HVDC) [32], and distributed generation [33], has been proposed to improve the performance in the VSC-HVDC system. However, these OPF approaches are developed without considering techno-economic parameters such as COE, NPC, and RF. Table 1 presents an overview of the reviewed literature.

Table 1
Comparative analysis of numerous configurations of off grid HMG systems in Egypt.

Ref	HMG set up	Site	area	Load demand	Objective			Tool	year
					NPC (k\$)	COE(\$/kWh)	RF (%)		
[5]	PV/WT/BAT/Conv	Tiba	City	Digestion plants	6894.7	0.1158	100	HOMER	2025
[6]	WT/FC/HT/BAT/PV/Elect	Cairo	City	School	1775.3	0.153	100	HOMER	2025
[7]	WT/Biomass/BAT/PV	Abu Minqar	Remote	village	3483.51	0.118	100	HOMER	2025
[8]	PV/WT/BAT/Conv	Sadat	Desert	Agricultural	194.01	0.12	100	HOMER	2022
[9]	PV/WT/BAT/Conv	Aswan	Desert	Households	853.63	0.25	100	HOMER	2024
[10]	PV/BAT/DG/Conv	Nakhl city	Urban	RO plant	502.66	0.107	93.1	HOMER	2020
[11]	PV/DG/WT/Conv/BAT	Abo ramada	Village	RO plant	3120	0.164	×	HOMER	2020
[12]	PV/DG/WT/Conv/BAT	Baltim	Coastal	RO plant	1048.04	0.10	95.5	HOMER	2021
[13]	PV/DG/WT/Conv/BAT	El Kharga Oasis	Desert	Households	×	0.23	×	IAOA	2022
[14]	PV/DG/WT/BAT/Conv	El Kharga Oasis	Desert	Village	103.4	0.108	90.1 %	HOMER	2024
[15]	PV/WT/FC/BAT/Conv	Safaga	Urban	House hold	123.29	0.15	100	HOMER	2020
[16]	PV/DG/WT/BAT/Conv	Marsa-Matruh	Community	House hold	351.22	0.22	75	HOMER	2020
[2]	PV/BAT/WT/Conv	Sinai	Remote	Tourism place	146.40	0.118	100	GWO	2021
[17]	PV/FC/ WT	Beni-suef	Remote	Village	2.69	0.45	×	FA	2020
[18]	PV/DG/ WT/ BAT/Conv	Qusier	Desert	Village	169.8	0.140	82.4 %	HOMER	2024
[19]	PV/FC/HT/Bio/WT/ELZ	Al-Aqaba	Remote	Village	6170	0.2106	×	MOA	2022
[20]	PV/WT/DG/BAT	Al-kharjah	Remote	Village	5798.7	0.1990	×	DCS	2024
[21]	PV/WT/BAT/MT/HT	Owienat	Desert	Residential	4228.11	×	×	HOMER	2024
[22]	PV/WT/BAT/DG/HT	Baris	Desert	Residential	1852.42	×	×	QIARO	2024
[23]	PV/BG	Sohag Al Gadida	Village	House hold	0.011	0.184	99.9	HOMER	2024
[24]	PV/ WT/ BAT	Noubarya	Village	Desalination	137.01	0.15	×	iHOGA	2018
[25]	PV/WT/biomass/HESS	Ataka	Desert	Households	8055.05	0.275	×	HBO	2023
[26]	PV/WT/DG/BAT/BIO/NGT	Quseer	Desert	Village	185.31	0.54	×	BOACA	2018
[27]	PV/WT/BAT/EV/HESS	Marsa-Matruh	Remote	Residential	13.06	×	×	AVOA	2024
[28]	PV/WT/DG/ BAT	Farafra	Desert	Domestic	187.18	0.213	90	IAOA	2023

1.3. Research gap and contribution

The HMG system has advanced considerably; however, several important issues remain unaddressed. Most current studies on HMG have concentrated on techno-economic assessment of households, rural villages, urban communities, and remote areas. Until now, there has been no comprehensive investigation into the role of HMG in the health clinics located in the desert areas of Egypt. Hence, health clinics in these sites have been significantly overlooked in the literature. Furthermore, health clinics in desert regions face unique challenges including critical continuous power demands, operation of sensitive medical equipment, and logistical limitations. Moreover, the COVID-19 pandemic has underscored the vital role of resilient energy infrastructure in healthcare facilities, particularly in desert areas and underserved areas where dependable grid power is limited or unavailable. A limited number of studies have performed detailed sensitivity analyses to examine how key parameters influence techno-economic metrics such as NPC, financial conditions, and the variability of RES availability.

This work addresses gaps in the literature by designing the techno-economics of an island HMG that can satisfy the load demand of the health clinic in the desert site of Egypt. The primary goal of this research is to decrease COE and NPC, and enhance LPSP and RF. This research combines mathematical modeling, resource and load characterization, optimization (using HOMER), sensitivity analysis, and multi-criteria techno-economic evaluation to design and assess the HMG system. This work offers the following key contributions:

1. Developing the HMG to satisfy the electricity needs in the Nubian Desert region of Aswan, while overcoming the challenges related to its optimal sizing and cost, leads to a more effective and economical energy solution.
2. Conducting the technical and economic performance analysis to evaluate the feasibility of four alternative microgrid options: WT/Conv/PV/BAT, PV/DG/BAT/Conv, PV/Conv/WT/BAT/DG, and DG/WT, provides valuable insights into the most efficient HMG setup.
3. Evaluating the sensitivity analysis based on fluctuations in wind speed and solar irradiance, cost of diesel, and discount rate parameters helps to achieve optimal, accurate, and useful results for the techno-economic analysis.

1.4. Organization of paper

The rest of the paper is organized as follows: Section 2 is divided into three subsections. The first subsection outlines the input parameters, including load profile, site description, wind speed, solar radiation, and diesel consumption. The second subsection details each component of the system, including the PV, DG, WT, BAT, and Conv. The last subsection describes the analysis of economic parameters and constraints consisting of COE, NPC, RF, and LPSP. Section 3 provides the simulation results, while Section 4 concludes the work.

2. Methodology

In this study, the electrical power needs of a health clinic in the Nubian (desert village of Aswan) are simulated through HOMER Pro. The simulation aims to determine the techno-economic configuration. The design process utilizing HOMER Pro is depicted in Fig. 1. The initial phase focuses on assessing the pre-feasibility, including environmental conditions, technical specifications, cost factors, and electrical power usage. In the subsequent phase, different HMG configurations are analyzed, and their techno-economic feasibility is evaluated individually. The power balancing calculations are carried out for every HMG over a year, with results summarized at one-minute intervals. In each iteration of the optimization process, the algorithm seeks the best values for various decision variables, including the number of DGs, battery capacity, size of PV systems, number of WTs, and converter ratings to minimize the

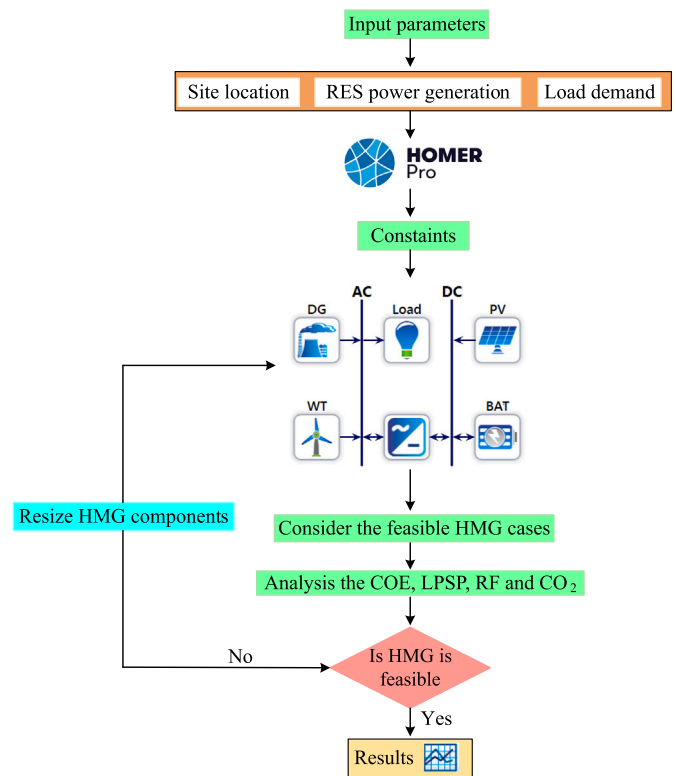


Fig. 1. The flow chart used to assess the optimal techno-economic of the designing of HMG system.

NPC in each optimization iteration. After evaluating all potential configurations and achieving convergence to the desired accuracy in the NPC, the algorithm produces a ranked list of viable solutions based on their economic performance. Furthermore, this section provides a brief description of the system input parameters, specifications of the proposed system components, and economic analysis.

2.1. Input parameters

Input parameters are used in the HOMER Pro to identify the optimal results for different system configurations. These factors include selected site location, solar radiation, electric load demand, diesel consumption, and wind speed.

2.1.1. Selected site for case study

In the paper, the Nubian village in Aswan is selected as a case study, as shown in Fig. 2. The proposed area is located in Upper Egypt at 24° 5' N latitude and 32° 53' E longitude and has the most sunshine in the country, with an average of 10.6 hours every day, which totals nearly

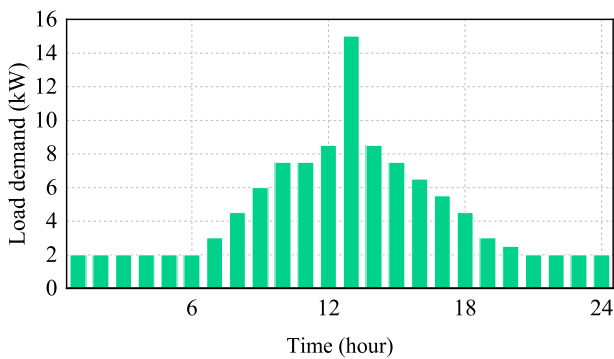


Fig. 2. Location of selected site under study on HOMER window map.

Table 2

The load demand of the proposed site based on the type of load and its operating conditions.

Device	Rating W	Quantity	Operating time hrs	Total energy kWh
CLFs lamps	24	6	10	1.44
Fluorescent tubes	40	8	12	3.84
Refrigerator (vaccine)	60	1	20	1.2
Refrigerator (blood bank)	60	1	20	1.2
Refrigerator (food)	400	1	18	7.2
Fan	50	10	20	10
AC	1500	4	12	72
ECG machine	100	1	2	0.2
Microscope	20	2	3	0.12
Surgical lights	500	1	2	1.0
Oxygen Concentrator	270	1	2	0.54
Medical exam table	300	1	2	0.6
Medical exam Chair	300	1	2	0.6
Ventilators	300	1	3	0.9
Operating Room Tables	200	1	3	0.6
Ultrasound machine	800	1	2	1.6
Desktop Computer	150	1	20	3
Radio Receiver	30	1	12	0.36
Mobile Charger	20	8	4	0.6
TV set	250	1	12	3
Total				110

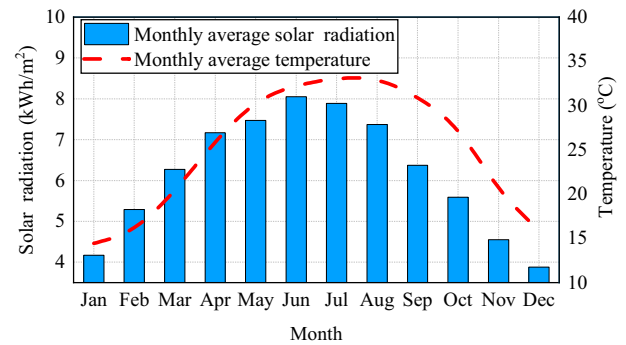
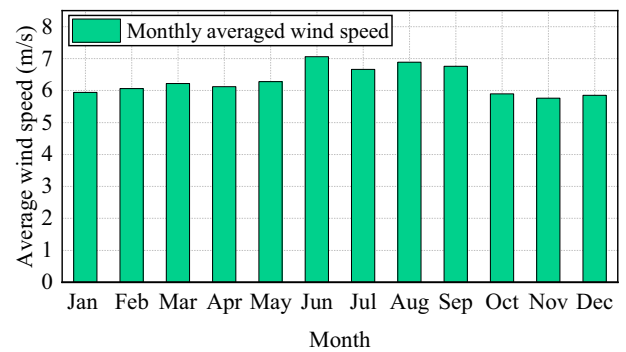
**Fig. 3.** Hourly health clinic load demand employed in HOMER.

3863 hours annually with nominal cloudy days. The subtropical desert temperature and vast desert regions make Aswan an ideal location for effective energy exploitation almost throughout the year.

2.1.2. Load profile

In this study, the electrical power demand of a health clinic in Nubian village is categorized by component type and its power ratings, quantity, operating time in hours, and corresponding energy consumption as given in Table 2. Based on this data, the daily power utilization is determined to be 110 kWh/day, with the peak load requirement of 15 kW. In the HOMER, the load was modeled with a minimum baseline of 2 kW and a maximum of 15 kW, based on hourly load demand illustrated in Fig. 3. This load profile represents the typical operating behavior of a rural medical clinic, showing higher demand during daytime working hrs and a continuous baseline at night time to sustain the load requirement.

The power distribution demand was developed based on the detailed equipment presented in Table 2. To accurately represent operational priorities, the load is classified into critical and non-critical categories. Critical load consists of blood bank and vaccine refrigerators, ventilators, oxygen concentrators, diagnostic equipment (such as ultrasound and ECG machines), and essential lighting which were designed with a higher priority. In contrast, non-critical load comprises administrative computers, televisions (TVs), fans, and mobile chargers, which were modeled with a lower priority.

**Fig. 4.** Monthly average solar radiation and temperature in Nubian.**Fig. 5.** Monthly average wind speed in Nubian.

2.1.3. Solar radiation

Solar energy data of Nubian collected from NASA indicates that this location experiences high levels of solar irradiation and temperature (°C) as shown in Fig. 4. It indicates a steady increase in solar radiation from January to its peak in June and then gradually declining until reaching the lowest level in December. June has the highest solar radiation at 8.05 kW per square meter (kWh/m²), while December has the lowest solar radiation value at 3.88 kWh/m². The total yearly maximum and minimum solar radiation are 2270.5 kWh/m² in June and 1955.6 kWh/m² in December, respectively, with an average solar irradiation of around 6.17 kWh/m². These variations in solar irradiation are caused by seasonal changes, with more intense sunlight during summer and less intense sunlight during winter. Similarly, the temperature in Aswan rises steadily from January to July, peaking in July. After July gradually decreases each month. The selected area's temperatures range from 14.43 to 33.02 °C with an average yearly temperature of approximately 24.9 °C.

2.1.4. Wind speed

Nubian's monthly wind speed average data is provided in Fig. 5. It reveals that consistent and low speeds mark the wind speed in the proposed area throughout the year. The mean wind speed fluctuates between 5.76 and 7.06 m/s, reaching its maximum of about 7.06 m/s in the summer and decreasing to approximately 5.76 m/s in the winter. Indeed, June recorded the maximum wind speed, with August coming second. Around 6.29 m/s is the average wind speed. This wind speed data offers detailed insights into the wind energy potential available in Nubian year around which can be utilized for wind power production.

The ambient temperature, solar irradiance, and wind speed data for Nubian (Aswan) were sourced from the NASA POWER database [34]. Wind speeds recorded at 23 meters (m) were adjusted to turbine hub height using a power-law relationship with an exponent of 0.143. Table 3 presents the monthly average GHI, wind speed at 23 m,

Table 3
Monthly average wind speed, solar radiation, and temperature in Nubian.

Month	Solar Radiation (kWh/m ² /day)	Temperature (°C)	Wind speed (m/s)
January	4.17	14.43	5.94
February	5.29	16.21	6.06
March	6.27	20.46	6.22
April	7.17	25.86	6.12
May	7.47	30.21	6.28
June	8.05	32.23	7.06
July	7.89	33.02	6.66
August	7.37	32.86	6.89
September	6.73	30.88	6.76
October	5.59	27.11	5.9
November	4.55	20.75	5.76
December	3.88	15.91	5.85

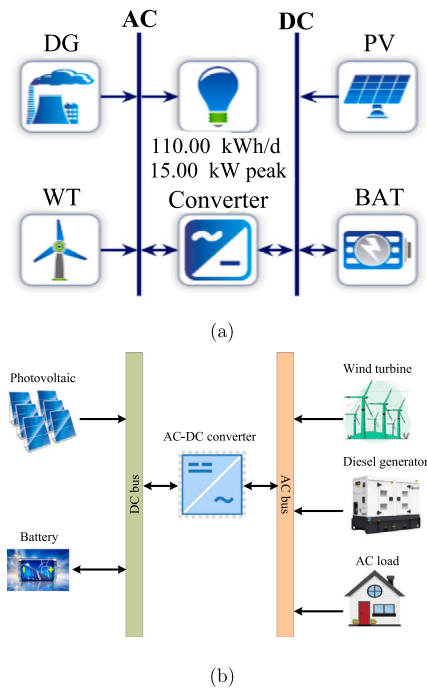


Fig. 6. Designed off-grid HMG system (a) HOMER model, and (b) schematic diagram.

and temperature, which served as input parameters for HOMER based HMG simulation.

2.1.5. Diesel

As a part of economic reforms in Egypt, the government has reduced financial subsidies on petroleum products. As a result, the cost of diesel in Egypt is 0.3 dollars per liter (\$/L) in the year 2025 [35].

2.2. Description of proposed system components and their specifications

The HOMER and schematic diagram of the developed HMG are shown in Fig. 6, in which DG, AC load, and WT are directly connected to AC bus. Whereas, PV and batteries employ the DC/AC converter to link with AC bus. The following sections will examine the developed HMG system to determine the optimal size for each power generator in relation to the specified objective functions based on the potential of RES.

2.2.1. Modeling of PV

Aswan benefits from excellent solar radiation throughout the year, making it a prime location for converting solar irradiation into electrical power. The available solar irradiation (S) and surface temperature (T_c)

Table 4
Technical and economic parameters utilized in the developed HMG.

Component	Description	Specification
PV [8]	PV Type	Canadian flat solar MaxPower
	Capital cost	0.4 k\$
	Operation & Maintenance cost	10 \$/kW/yr
	Lifetime	25 yrs
	Nominal operating cell temperature	44.20
	Rated Power	0.325 kW
	Temperature coefficient	-0.4294
	Derating factor	85 %
	efficiency of STC	13 %
	Replacement cost	400 \$/kW
WT [14]	WT Type	Eocycle(E010) model
	Capital cost	11,000 \$/turbine
	Operation & Maintenance cost	30 \$/kW/yr
	Replacement cost	10,000 \$/kW
	Cut in wind speed	2.75 m/s
	Rated power	10 kW
	Life time	20 yrs
	Rated velocity	6.5 m/s
DG [35,42]	Cut out wind speed	20 m/s
	Capital cost	1000 \$/kW
	Operation & Maintenance cost	0.02 \$/op.hr /yr
	Replacement cost	1000 \$/kW
	Minimum load ratio	25 %
	Fuel curve slope	0.273 L/hr/kW output
	diesel fuel price	0.3 \$/L
	Lifetime	15,000 hrs
BAT [43]	Capital cost	0.3 k\$
	Operation & Maintenance cost	5 \$/bat/yr
	Replacement cost	0.15 k\$
	Maximum rating	83.4 Ah
	Nominal capacity	1 kWh
	Nominal voltage	12 V
	Maximum charge current	16.7 A
	String size	4
	Highest discharge current	24.3 A
	SOC range	20 100 %
Converter [44]	Lifetime data transfer rate.	504.9 kWh
	Initial investment cost	200 \$/kW
	Operation & Maintenance cost	0 \$/kW/yr
	Converter's efficiency	95 %
	useful life	15 yrs
	Relative capacity	100 %
	Replacement cost	200 \$/kW

at time (t) determine each PV module's power generation, which is calculated using the following formulas [36].

$$P_{PV}(t) = PV_{rated} \times \frac{S(t)}{S_{ref}} [1 + K_T (T_C - T_{ref})] \quad (1)$$

$$T_C(t) = T_a(t) + \frac{NOCT - 20}{800} \times S \quad (2)$$

where $P_{PV}(t)$ and PV_{rated} are the output and rated powers of PV panel, $S(t)$ and S_{ref} represent the solar radiation and its reference value. K_T is the temperature coefficient, $T_a(t)$ and $NOCT$ are the ambient temperature ($^{\circ}C$) at the time interval (t) of the selected site and normal operating cell temperature. The $NOCT$ is the cell temperature of a PV module operating at 1000 watts per square meter (W/m^2) of solar radiation and $20^{\circ}C$ ambient temperature. $NOCT$ typically varies from $42^{\circ}C$ to $46^{\circ}C$. In this study, MaxPower (CS6X-325 P) flat type PV panel is employed. The technical specifications and cost details of the selected PV model are given in Table 4.

2.2.2. Modeling of WT

Wind turbines produce electrical power from kinetic energy when the wind speed is within the cut-in and cut-out limits. For an accurate

WT modeling, three primary aspects must be considered: tower height, the power output curve of WT, and data of wind speed from the selected site. The power generated by WT can be calculated by Eq (3) [37].

$$P_{WT} = \begin{cases} P_{rated}^{WT} \times \left(\frac{V - V_{in}}{V_r - V_{in}} \right)^3 & V_{cutin} \leq V \leq V_r \\ P_{rated}^{WT} & V_r \leq V \leq V_{cutin} \\ 0 & V \geq V_{out}, V \leq V_{cutin} \end{cases} \quad (3)$$

where, P_{WT} and P_{rated}^{WT} are the output and rated powers of WT, V_{cutin} and V_{out} represent the cut in and cut off wind speeds, and the rated wind speed is denoted by V_r . Eq (3) defines the WT's power curve behavior, showing the performance pattern for output power generation. The WT's output power is zero when wind speeds are below the V_{cutin} threshold or above the V_{out} limit. However, in the range from V_{cutin} to V_r , the produced power rises approximately with the cube of the wind speed. Beyond the V_r and up to the V_{out} , the WT sustains a steady rated power output. Eq (4) defines the specific hub height (V_{WT}), which is related to the reference value of wind speed (V_{ref}) for a wind turbine at a given location.

$$V_{WT}(t) = V_{ref}(t) \times \left(\frac{h_{WT}}{h_{ref}} \right)^\alpha \quad (4)$$

where, h_{WT} and h_{ref} are the WT hub height and the reference point of height at which the anemometer is employed. α represents the coefficient of friction. The friction coefficient depends on several factors: height above ground, wind speed, time of year, and roughness of the terrain. This coefficient typically has a value of 0.14 [38]. In this work, the WES-100 model (Ecocycle type) WT type is used. Table 4 outlines an overview of its technical specifications and cost details.

2.2.3. Diesel generator

In this proposed HMG system, DG is used as the backup power generation alternative, and supplies power to meet the load requirement under the insufficient power generation from WT and PV power sources or battery consumption during periods of high power demand. The fuel consumption liter per hour (L/h) of DG is dependent upon output power and can be determined as [39]:

$$F_C = f_{g,r} \times P_r + f_{go} + P_r(t) \quad (5)$$

where, F_C , $f_{g,r}$, f_{go} , and P_r are the fuel consumption, DG fuel curve intercept coefficient, DG fuel curve slope, and rated capacity of the utilized DG. Technical and cost details for DG employed in this research are given in Table 4.

2.2.4. Battery

In the developed HMG setup, the battery is utilized as a BSS. Battery helps to satisfy the power needs during extended periods of reduced PV or WT availability such as during night time or calm weather. This ensures that the load demand is maintained and the utilization of RES is maximized. The nominal capacity of the battery can be calculated using Eq (6). Additionally, the energy stored in the battery at time (t) can be calculated through Eq (7) for battery charging and Eq (8) for battery discharging [40].

$$C_b = \frac{E_L AD}{\eta_{inv} \times \eta_{BAT} \times DOD_{max}} \quad (6)$$

$$E_{BAT}(t) = (1 - \sigma) \times E_{BAT}(t-1) + \left[\left(P_g(t) - \frac{P_L(t)}{\eta_{Conv}} \times \eta_{BAT} \right) \right] \times \Delta t \quad (7)$$

$$E_{BAT}(t) = (1 - \sigma) \times E_{BAT}(t-1) - \left[\frac{P_L(t)}{\eta_{Conv}} - \frac{P_g(t)}{\eta_{BAT}} \right] \times \Delta t \quad (8)$$

where C_b , E_L , and AD are the capacity of the battery, load demand, and autonomy days, respectively. DOD_{max} is the maximum depth of

discharge of the battery. η_{Conv} and η_{BAT} are the efficiencies of inverter and battery. E_{BAT} is the battery power. P_g and P_L provide the information about the power generation and power load demand. In this study, BAT-Trojan 75 battery type is employed, and its technical and economical specifications are given in Table 4.

2.2.5. Power converter

In this proposed system, DC power is generated by PV, while WT and DG generate AC power. To supply power to AC loads, a power converter is employed to connect the DC and AC buses. The size of the converter (P_{conv}) is determined by maximum load demand $P_{peak}(t)$ and efficiency of the inverter (η_{inv}) as specified in Eq (9) [41].

$$P_{conv}(t) = \frac{P_{peak}(t)}{\eta_{inv}} \quad (9)$$

The power converter's capacity can be expressed as the amount of electrical power transferred in both directions. The efficiency of the power converter designed for this study can be determined through Eqs (10) and (11) [15].

$$\eta_{DC/AC}^{conv} = [P_{PV}(t) + P_{BAT}(t)] \leq P_{conv} \quad (10)$$

$$\eta_{AC/DC}^{conv} = [P_{WT}(t) + P_{DG}(t)] \leq P_{conv} \quad (11)$$

where, $\eta_{DC/AC}^{conv}$ and $\eta_{AC/DC}^{conv}$ are the efficiencies of power conversion from DC to AC and AC to DC. The cost and other parameters for the converter used in this study are shown in Table 4.

2.3. Objective function

This research selects COE and LPSP parameters as main objective functions, with the goal of optimizing an HMG system that provides a reliable energy supply at the lowest possible cost.

2.3.1. Cost of electricity

COE includes capital, operation and maintenance (O&M), and replacement costs. HMG systems typically have minimal fuel expenses, leading to reduced O&M costs, though their initial capital investments are generally high. This work considers the operational costs for the HMG configuration consisting of expenses for the PV, DG, WT, BAT, and Conv components. The COE represents the average expense of generating useful electrical power by the system, measured in \$/kWh, and is determined using Eq (12) [45].

$$COE = \frac{TNPC}{\sum_{h=1}^{h=8760}} P_{load} \times CRF \quad (12)$$

$$TNPC = NPC_{cap} + NPC_{O\&M} + NPC_{rep} \quad (13)$$

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (14)$$

where, CRF, TNPC and P_{load} describe the capital recovery factor, total net present cost, and load power. NPC_{rep} , $NPC_{O\&M}$, and NPC_{cap} represent the replacement, operation and maintenance, and capital net present costs. n and i describe the number of years and the rate of interest.

2.3.2. Loss of power supply probability

LPSP is the probability that the system fails to satisfy the load requirement. This failure may arise from technical issues or insufficient renewable power generation input [46]. In this work, BSS is used to guarantee reliable operation. The LPSP is evaluated using a chronological method [47] over the time from 0 to T, as described in Eq (15). To optimize system efficiency, it is important to minimize excess energy and identify the appropriate BSS configuration. This allows for

uninterrupted power generation in the absence of alternative power sources, maintaining sufficiency.

$$LPSP = \frac{\sum (P_{load} - P_{PV} - P_{WT} + P_{DG} \pm P_{SOCmin})}{\sum P_{Load}} \quad (15)$$

where, P_{DG} , P_{PV} , and P_{WT} are the total power generated by DG, PV, and WT, respectively. P_{load} is the power of load demand. Furthermore, the following condition is applied in the assessment of the developed system's reliability, as discussed in Eq (16).

$$P(t)_{production} \geq P(t)_{load\ demand} \quad (16)$$

2.3.3. Renewable fraction

RF indicates the proportion of RES used to satisfy load demand. A 100 % RF implies that RES entirely meets the load [48]. The RF can be determined through Eq (17).

$$RF = \left(1 - \frac{\sum P_{diesel}}{\sum P_{PV} + P_{WT}} \right) \times 100 \quad (17)$$

As stated in Eq (17), an RF value close to 1 represents the maximum power production from RES, while values of zero reflect the minimal contribution from RES. For the proposed configuration, a higher RF is preferable, as it ensures more power is generated by RES.

2.3.4. Carbon emission

Carbon emission (CE) refers to the total amount of carbon dioxide released per kg as a result of the project [11], which is directly related to the quantity of fuel consumed.

$$CE = \sum_{t \in T} TCO_2 \times E_{diesel} \quad (18)$$

where, TCO_2 and E_{diesel} are the total of CO_2 emission tons per kilowatt hour (Ton/kWh) and total fuel utilized by DG (kWh). 1 L of DG fuel will produce 2.68 kg CO_2 according to the IPCC guidelines.

3. Results and discussion

This section provides the key capabilities of different HMG components, offering an in-depth assessment of the technical and economic aspects of the proposed HMG system. The study includes techno-economic simulations for multiple possible options, including DG/Conv/BAT/PV, WT/DG, WT/PV/Conv/BAT, and WT/DG/PV/BAT/Conv. Using HOMER software, power balances are calculated at each time step to simulate the performance of each configuration. It determines the optimal techno-economic setup based on its ability to satisfy power requirements under given conditions, as provided in Fig. 1. Moreover, HOMER provides a detailed cost analysis for 25 years of project duration, ranking potential scenarios and evaluating them based on their NPC, RF, and LPSP factors. The following subsection examines and analyzes four HMG schemes derived from HOMER's hourly simulation.

3.1. Scenario: WT/PV/BAT/Conv

In this scenario, the HMG system incorporates WT and PV RES to fulfill the proposed load requirement of the chosen site Nubian (Aswan). A BSS is also integrated to store excess power when production surpasses load demand. The batteries ensure a continuous power supply when RES based power generation is insufficient. The system configuration includes 30 kW of PV, one (EO10) WT, 116 batteries (there are 29 parallel strings, each containing 4 batteries), and a 16 kW Conv. The PV/WT/BAT/Converter setup results in an NPC of 78.2 k\$, and a COE of 0.165 \$/kWh, with a detailed analysis of cost presented in Fig. 7.

Fig. 7, demonstrates that the largest share of NPC is contributed by batteries, with WT and PV ranking 2nd and 3rd respectively. The annual power production from WT and PV is given in Fig. 8. About 47 % of

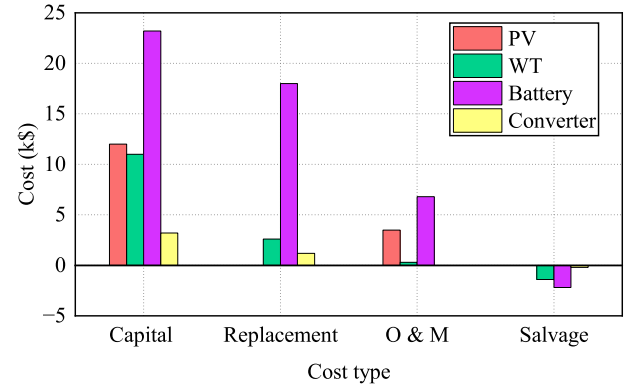


Fig. 7. An overview of the expenses for the HMG scenario 1.

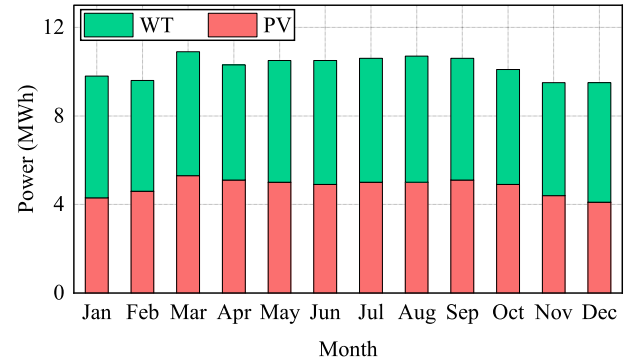


Fig. 8. Overview of the annual power output from WT and PV in Case 1.

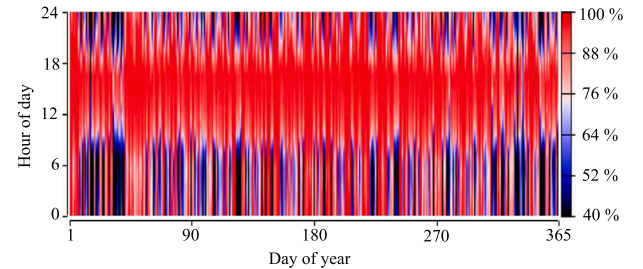


Fig. 9. An overview of the annual power output of batteries in scenario 1.

the total power is supplied by the PV, while WT accounts for the remaining 53 %. The balance of power generation between these RES is crucial for boosting the reliability and resilience of the HMG, particularly in response to deviations in weather and RES accessibility. This setup minimizes reliance on any single RES, thereby improving the system's reliability in case of weather-related disruptions or equipment malfunctions.

PV generally produces electricity during the day hrs when solar irradiation is available, while WT generates power when wind speeds are adequate. By combining both WT and PV, the chance of having a continuous power supply improves, as weather conditions are often unpredictable. This combination of RES contributes to a more stable balance of power between supply and demand within HMG system, ensuring a more consistent electricity supply during extreme weather events such as storms or heatwaves.

Fig. 9 illustrates the performance of batteries with their state of charge (SOC). SOC offers real time insights into the battery's charge level, facilitating efficient energy distribution and preventing overcharging and deep discharging. Additionally, accurate SOC tracking ensures

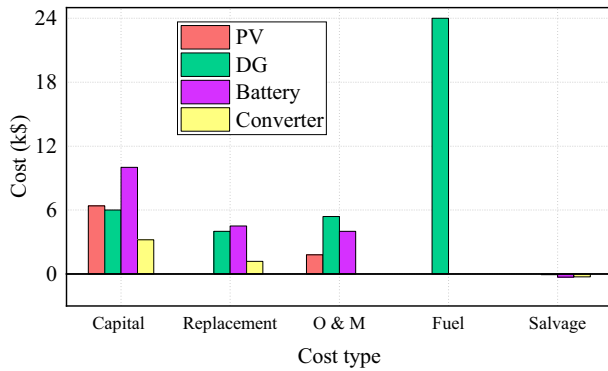


Fig. 10. Cost summary of HMG configuration in case 2.

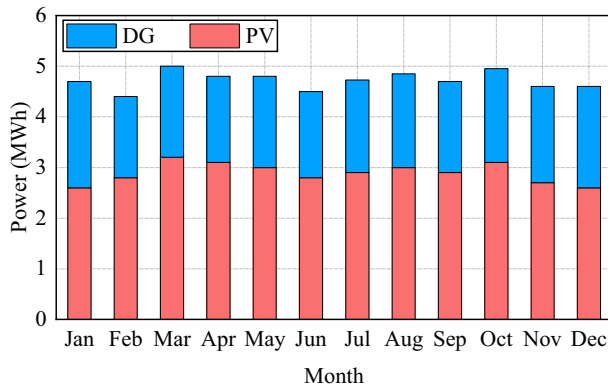


Fig. 11. Overview of the annual power output from DG and PV in Case 2.

that HMG maintains enough energy storage to meet demand in the case of low output from PV and WT. Consequently, SOC determines when the battery should discharge to export power to the load when WT and PV power generation are insufficient. This HMG arrangement suffers from 0.059 % of unmet load demand, and provides 10,660 kWh/yr of renewable energy curtailment with 100 % of RF.

3.2. Scenario 2: PV/DG/BAT/Conv

In this scenario, PV is combined with BSS to satisfy the load requirements in the desert area of Aswan. When PV generation is high, it supplies power to the load and charges the BSS. During periods when PV output is low, DG steps in to meet the load demand. The integration of batteries further boosts the system's reliability by compensating for any unsatisfied load. In this case, the techno-economic configuration consists of 6 kW of DG, 16 kW of PV, 36 lead acid batteries, and 16 kW of Conv. The suggested HMG scheme has NPC of 66.27 k\$ and COE of 0.140 \$/kWh. Fig. 10 provides a detailed cost assessment.

Fig. 10 indicates that the DG contributes the most to the NPC, with the batteries and PV following in sequence. Fig. 11 summarizes the electricity production from DG and PV. The DG accounts for 38.7 % of the total power output, while PV provides the remaining 61.3 %. Fig. 12 highlights the performance of batteries in the DG/PV/Conv/BAT setup, depicting the scenario when BSS transfers power to the load, and consumes power from PV and DG. Meanwhile, Fig. 13 displays fuel consumption by DG in the proposed configuration. In this setup, DG operates for 3875 hours per year (h/yr), producing CO₂ emissions of 18,025 kg/yr. The corresponding fuel consumption is approximately 6725 liters per year (L/yr). As a result, substantial fuel expenses are the primary limitation in this HMG system. Furthermore, this HMG arrangement experiences 0.0340 % of unmet load demand, 44.2 % of RF, and results in 5157 kWh/yr of energy curtailment.

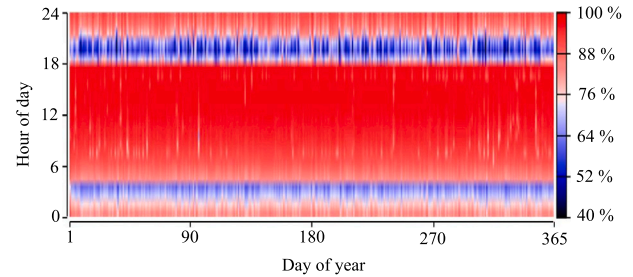


Fig. 12. An overview of the annual power output of batteries in scenario 2.

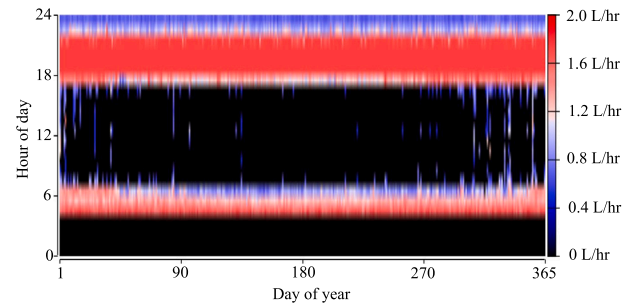


Fig. 13. Overview of the annual diesel used by DG in scenario 2.

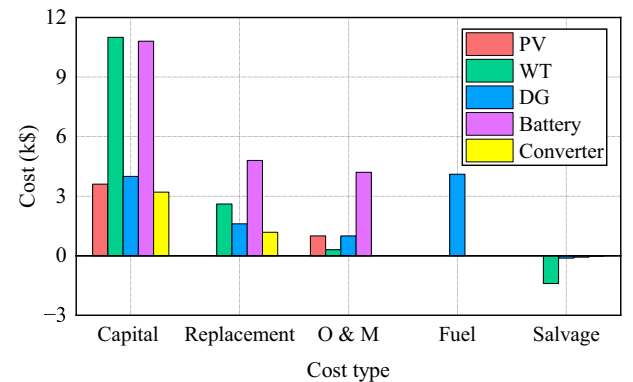


Fig. 14. An overview of the expenses for the HMG scenario 3.

3.3. Scenario 3: DG/WT/PV/BAT/Conv

In scenario 3, WT and PV are combined with DG and BSS to meet the load requirement in the selected site Nubian. In this HMG scheme, WT and PV power generation sources are used to satisfy the power demand and charge battery during the times of high power output. However, DG is utilized to fulfill the load need when the output power from WT and PV is low. Similarly, BSS enhances the system's reliability by addressing any remaining unmet load needs. During this condition, the 1 E010 WTs, 9 kW of PV, 4 kW of DG, 16 kW of Conv, and 36 batteries form the most effective techno-economic setup. This developed configuration has NPC of 51.76 k\$ and COE of 0.107 \$/kWh. The comprehensive cost summary analysis is shown in Fig. 14. Fig. 14 illustrates that BSS provided the highest percentage of the NPC, followed by WT and DG.

Fig. 15 presents the electrical power output from DG, WT, and PV. DG contributes 3.9 %, WT generates 76.1 %, and PV supplies 20 % of the total electricity generation. The performance of the BSS is illustrated in Fig. 16, highlighting the situation where BSS supplies power to the load, and consumes power from DG, PV, and WT sources. Fig. 17 depicts the fuel utilization by DG in the developed HMG system. In this

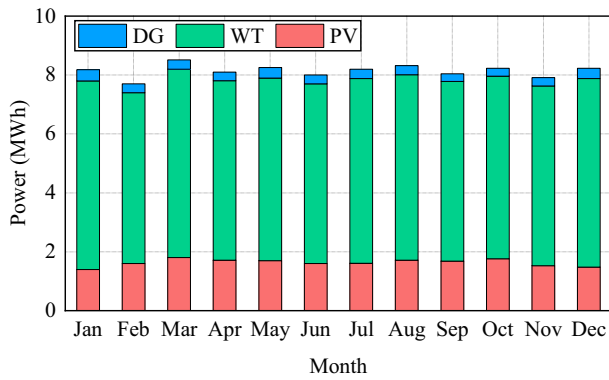


Fig. 15. An overview of the annual power output from DG, WT, and PV in Case 3.

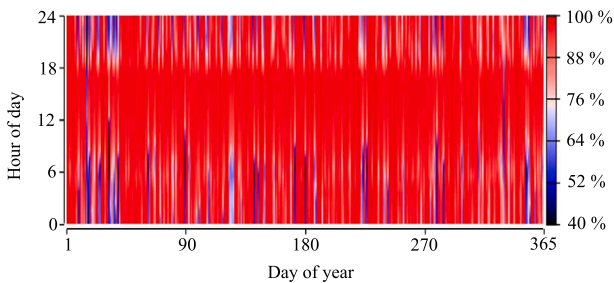


Fig. 16. An overview of the annual power output of batteries in scenario 3.

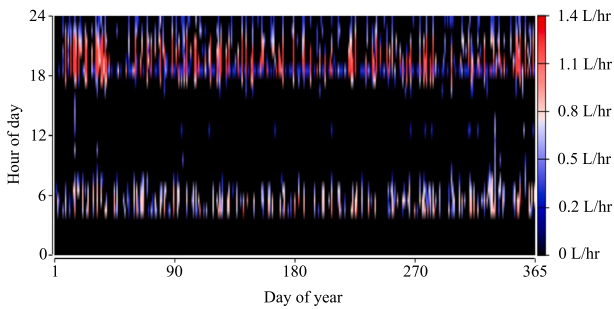


Fig. 17. An overview of the annual fuel utilization by DG scenario 3.

configuration, DG operates for 1106 h/yr, generating 3051 kg/yr of CO₂ emissions. The associated fuel utilization is around 1139 L/yr. Moreover, this HMG system records 90.7 % of RF, 0.0003 % of unmet load demand, and produces about 250 kWh/yr of excess energy.

3.4. Scenario 4: DG/WT

In case 4, WT and DG provide power directly to the load without requiring power converter and BSS. As both the load requirement and power generation are in AC form, there is no need for a power converter and BSS in this situation. The techno-economic evaluation of this HMG demonstrates that the optimal setup consists of 3 EO10 WTs and 12 kW of DG. This configuration has 70.809 k\$ of NPC and COE of 0.150 \$/kWh. A comprehensive breakdown of the costs is provided in Fig. 17. WT contributed the largest proportion of the NPC, followed by DG, as illustrated in Fig. 18.

The power outputs from the DG and WT are illustrated in Fig. 19, with the DG contributing 2.7 % and the WT generating 97.3 % of the total electricity. Fig. 20 shows the fuel consumption by DG in this HMG configuration. In this HMG scheme, the DG operates for 2175 h/yr, emitting approximately 6034 kg/yr of CO₂ emissions. The corresponding

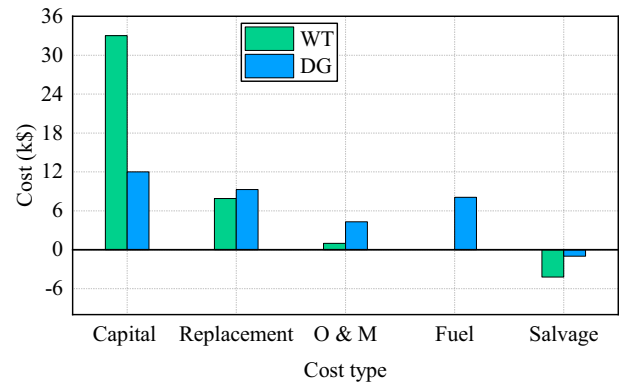


Fig. 18. Cost summary of HMG setup in Scenario 4.

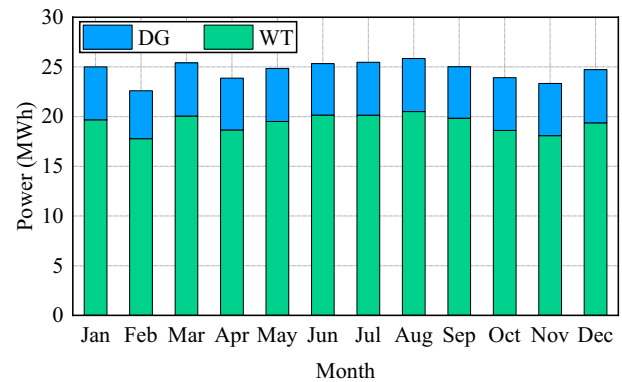


Fig. 19. An overview of the annual power output from WT and DG in scenario 4.

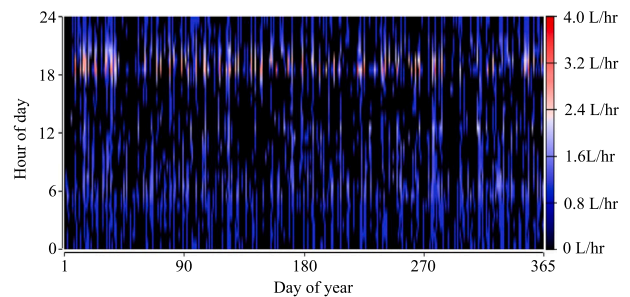


Fig. 20. An overview of the annual fuel utilization by DG scenario 4.

fuel consumption is approximately 2251 L/yr. Additionally, the system achieves an RF of 81.1 %, experiences only 0.0118 % unmet load demand, and results in around 8298 kWh/yr surplus energy.

3.5. Comparison among HMG configurations

A comparative analysis of the cost and environmental parameters examines the techno-economic factors of the four HMG schemes discussed in this section. To ensure a detailed and clear comparison Table 5, and Fig. 21 illustrate the results for various cost components of the evaluated HMG, as well as the final cost economic ranking based on NPC.

Table 5 clearly indicates that HMG based on the WT/PV/BAT/Conv shows the highest NPC. The costs associated with this system are 49.4 k\$, 28 k\$, 21.78 k\$, and 3.8 k\$ for capital, replacement, O & M, and salvage respectively. Additionally, PV/DG/WT/Conv/BAT combination in the HMG is considered an efficient techno-economic option meeting the load demands in the desert area of Aswan, offering the lowest

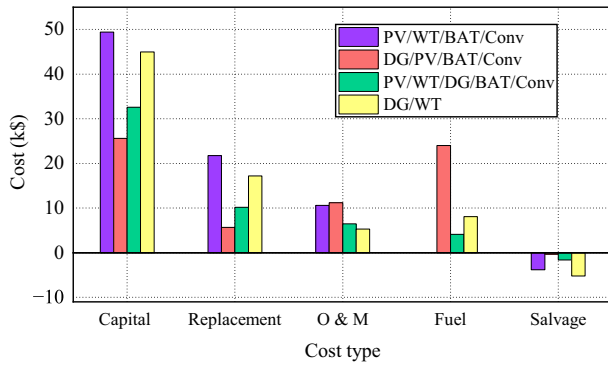


Fig. 21. An overview of cost comparison among different HMG configurations.

O & M, capital, and replacement costs of 10.18 k\$, 32.6 k\$, and 16 k\$ respectively. In Table 5, the final analysis of COE reveals that the PV/BAT/WT/Conv configuration has the highest COE at 0.165 \$/kWh, whereas the WT/DG/PV/BAT/Conv setup has the lowest COE at 0.107 \$/kWh. This demonstrates how the designed HMG delivers the most cost effective electricity to meet the load requirements. Furthermore, Table 5 verifies that, in the proposed HMG, the DG operates for approximately 1106 h/yr, resulting in a yearly CO₂ emission of 3051 kg. Using the stated emission factor of 2.68 kg CO₂/L of diesel, the corresponding diesel consumption is calculated as 1139 L/yr. The reduced DG operation indicates that the integrated PV and WT system effectively meets the majority of energy demand, minimizing diesel fuel dependence and associated emissions. The system exhibits negligible unmet load demand, ensuring that the clinic's critical loads are continuously supplied without power interruption. Additionally, a small portion of power generation is curtailed as an energy surplus. In contrast, the WT/DG and PV/DG/BAT/Conv systems exhibit higher DG runtime of 2175 h/yr and 3875 h/yr, leading to annual CO₂ emissions of 6034 kg and 18,025 kg respectively. Therefore, the developed HMG shows strong performance highlighting its effectiveness in enhancing environmental sustainability and clinic power reliability, ensuring uninterrupted operation of critical medical equipment in remote areas. The PV/DG/Con/BAT is placed in second position and WT/DG is ranked in third place.

3.6. Sensitivity analysis

After designing the techno-economic HMG to fulfill load needs, while considering accessible energy sources and financial conditions, several system parameters may change. These parameters consist of variations in power generation from RES such as solar radiation and wind speed, as well as economic factors including the discount rate, component costs, and cost of diesel. Fluctuations in these parameters result from changes

Table 6
Sensitivity factors of the developed HMG setup.

Sensitivity parameter	Reference values	Limit
Wind speed	5.05	± 20 %
Cost of diesel	0.24	0.1, 0.3, 1
Solar radiation	6.54	± 20 %
Discount rate	3, 6, 9	3 to 9 %

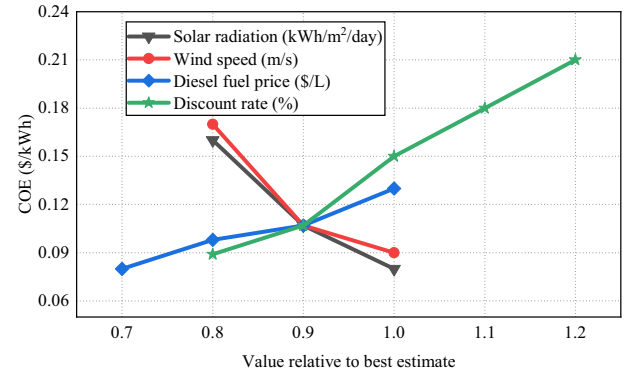


Fig. 22. The effect of sensitivity changes on the COE of the proposed HMG setup.

in data collection accuracy, timing, and policy shifts, all of which influence the system's actual design and operation. Therefore, performing a sensitivity analysis becomes crucial. In this study, the sensitivity evaluation was performed to assess the impact of varying factors such as discount rate, wind speed, price of diesel, and solar radiation. Changes in solar radiation and wind speed have a direct effect on the techno-economics of PV and WT. As a result, ± 20 % deviation in wind speed and solar radiation mean values are applied. The simulation was conducted using the established cost of diesel, but this price is expected to increase throughout the project's duration. Future price increases are anticipated when government subsidies for diesel are reduced. Moreover, deviations in global diesel prices and shifts in global export and import worldwide further influence diesel expenses. In this analysis, diesel prices of 0.1, 0.3, and 1.0 \$/L along with discount rates of 3, 6, and 9 % are employed to perform the sensitivity analysis. Table 6 presents selected values for this sensitivity assessment, along with their influence on the COE and NPC effectiveness on the designed HMG as shown in Figs. 22 and 23, respectively.

Fluctuations in solar irradiation and wind speed have a significant influence on the COE and NPC, as shown in Figs. 22 and 23. A reduction of 20 % in solar irradiation and wind speed leads to substantial increases in NPC, rising by 12 % and 17 % respectively. This rise is primarily driven by extended operation hrs and higher fuel utilization of

Table 5
An overview regarding the cost and environmental analysis among different configurations of the MG system.

Parameters	Unit	HMG configurations			
		PV/WT/BAT/Conv	DG/PV/BAT/Conv	PV/WT/DG/BAT/Conv	WT/DG
Capital cost	k\$	49.4	25.6	32.6	45
Replacement cost	k\$	28	22.2	16	26.6
O & M cost	k\$	21.78	5.68	10.18	17.2
Fuel	k\$	0	24	4.1	8.1
Salvage	k\$	3.8	0.34	1.62	5.2
COE	\$/kWh	0.165	0.140	0.107	0.150
NPC	k\$	78.22	66.27	51.76	70.4
CO ₂ emission	(kg/yr)	0	18,025	3051	6034
Unmet load demand	%	0.0596	0.0340	0.0003	0.0118
DG run time	hrs/yr	0	3875	1320	2175
RF	%	100	44.2	90.7	81.1
Curtailed	kWh/yr	10,660	5157	1250	8298
Rank	—	4	2	1	3

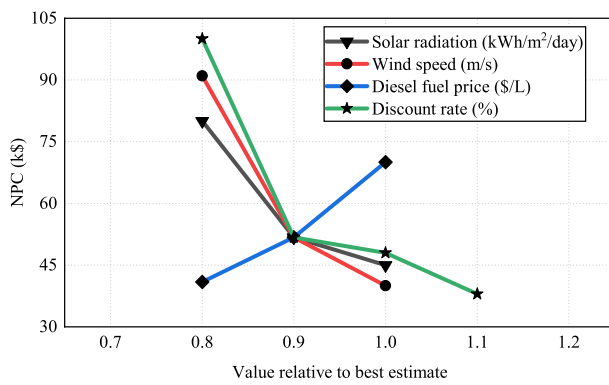


Fig. 23. The effect of sensitivity changes on the NPC of the designed HMG setup.

Table 7

Comparison analysis of the current study with similar research in Egypt.

Ref	Location	COE(\$/kWh)	NPC (k\$)	RF (%)	year
[14]	El Kharga Oasis	0.108	103.4	90.1	2024
[20]	Al-kharjah	0.199	5798.7	×	2024
[18]	Qusier	0.140	169.8	82.4	2024
[28]	Farafra	0.213	187.18	90	2023
[16]	Marsa-Matruh	0.22	351.22	75	2020
[11]	Abo ramada	0.164	3120	×	2020
[13]	El Kharga Oasis	0.23	×	×	2022
Present study	Nubian	0.107	51.76	90.7	—

DG, which is needed to offset the lower power production from PV and WT. As solar irradiation and wind speed increase, there is a corresponding reduction in both COE and NPC. This assessment demonstrates that the 20 % increase in wind speed and solar irradiation brings NPC down from 76.2 k\$ to 51.76 k\$, whereas COE decreases from 0.161 to 0.107 \$/kWh. As discount rates increase, NPC and the initial cost and NPCs of the system decrease because of a reduction in the BSS and the number of WT. Specifically, the lowest 51.76 k\$ of NPC is reached with a 9 % of discount rate, reflecting a 19 % decrease compared to the baseline at 6 %. This reduction is primarily attributed to the expensive replacement and initial costs of BSS and WTs, which result in a decrease in their quantities as discount rates increase. As a result, increasing the discount rate generally causes the COE to decrease. At a discount rate of 9 %, the COE increases to 0.12 \$/kWh, which is approximately 29 % of the initial rate of 3 %. This increase is largely attributed to higher O & M cost, resulting from increased fuel utilization and running of DG for longer hrs to offset the fewer WTs. However, even with a significant increment in diesel prices, the impact on COE and NPC remains minimal. When diesel price hits its most optimal point of 0.3 \$/L, the COE and NPC only rise by about 1.4 % and 1.6 % respectively.

In summary, the sensitivity analysis indicates that the changes in discount rate and weather patterns have a substantial effect on the techno-economic performance of HMG. As a result, these factors must be carefully considered during the project planning phase before making investment decisions. However, fluctuations in diesel prices are less critical when developing HMG project. The key reason for diesel investment is maintaining the reliability of the system, while adhering to the established limits aimed at reducing harmful gas emissions.

3.7. Comparison analysis with related studies

Comparing the current study with other works is difficult because of differences in system configuration, types of RES, load patterns, and component diversity. The validation of this research is based on evaluating the techno-economic aspects, including NPC, COE, and RF. Table 7 provides a comparison of the present study with similar research carried out in Egypt.

In previous works on similar proposed HMG systems, as shown in Table 7, reported the higher values of NPC and COE, and lower values of RF, as compared with the current proposed study. Furthermore, Table 5 confirms that the designed DG/WT/PV/BAT/Conv scheme is the most cost effective compared to the WT/Conv/PV/BAT, PV/DG/BAT/Conv, and DG/WT arrangements.

4. Conclusion

This study conducted the techno-economic assessment of HMG system to meet power demand of a health clinic with a daily energy requirement of 110 kWh, located in the desert area of Nubian in Aswan, Egypt. This analysis was conducted using HOMER Pro, with the system's viability assessed based on NPC, RF, COE and CO₂ emissions of the systems. Various HMG configurations were examined, including PV/DG/BAT/Conv, PV/WT/BAT/Conv, WT/DG, and PV/WT/BAT/Conv ensuring for an equitable comparison among options. Furthermore, sensitivity analysis was conducted to investigate the impact of variations in solar radiation, wind speed, discount rate, and diesel cost input parameters on the optimal solution.

The simulation results confirm that the proposed off grid HMG consisting of PV/DG/WT/BAT/Conv effectively satisfies the yearly and daily power demands of the health clinic in the Aswan with 0 % of unmet load demand. The optimal configuration of the designed HMG includes 4 kW of DG, 10 kW of WT, 9 kW of PV, 36 batteries, and 16 kW of Conv. This configuration achieves a COE of 0.107 \$/kWh, NPC of 51.76 k\$, and an RF of 90.7 %. Additionally, the designed HMG produces 3051 kg/year of CO₂ emissions, representing 14.47 % of the CO₂ emissions generated by DG/PV/BAT/Con (18,025 kg/year) and 33.5 % of those from a DG/WT setup (6034 kg/yr). Hence, the proposed HMG provides the cost effective, reliable, and sustainable solution for powering rural health care facilities.

CRedit authorship contribution statement

Asif Raza: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Zi-Hong Jiang:** Supervision, Investigation, Formal analysis, Conceptualization. **Yi-Die Ye:** Software, Methodology. **M Punhal Sahto:** Writing – original draft, Formal analysis. **Ahmed Lotfy Haridy:** Writing – review & editing, Writing – original draft. **Said I. Abouzeid:** Writing – original draft, Investigation. **Ghalib Raza:** Visualization, Resources. **Jibran Hussain:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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